

V.N

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CONFIDENTIAL

MARKING SCHEME

JUNE 2018

MATHEMATICS

9164/2

Section A: Pure Mathematics

1 (a) $\tan \widehat{AOB} = \frac{x}{\frac{x}{2}} = 2$ M1

$\widehat{AOB} = 1.107$ radians *not degrees and to 3 d.p.* A1 [2]

(b) $\widehat{BOC} = \pi - 2 \times 1.107$
 $= 0.9276$ radians B1

radius OB = 5.591 B1

Shaded area = $\frac{1}{2}r^2(\theta - \sin \theta)$
 $= \frac{1}{2} \times 5.591^2 [0.9276 - \sin 0.9276]$ M1

$= 1.990$ *or 1.991* A1 [4]

2 $\log_2(8y) = x + 5$

$2^{x+5} = 8y$

$2^{x+5-3} = y$

$2^{x+2} = y$

$4^x = 3y \Rightarrow 2^{2x} = 3(2^{x+2})$

$\Rightarrow 2^{x-2} = 3$

$(x-2)\ln 2 = \ln 3$

$x = 2 + \frac{\ln 3}{\ln 2}$

$= 3.5849$ *or 3.6. $\frac{\log 12}{\log 2}$* A1

$y = \frac{1}{3}(4^{3.5849})$ M1

$y = 48$ *= 47.997* A1

[7]

Link bth indices and log M1 *for laws of ind.*

A1

M1 *Substituting for*

for introducing logarithms M1

A1

M1

A1

$$3 \quad y^3 + 3xy^2 - x^3 = 3$$

$$3y^2 \frac{dy}{dx} + 3y^2 + 6xy \frac{dy}{dx} - 3x^2 = 0$$

M1

$$\frac{dy}{dx} = \frac{3x^2 - 3y^2}{6xy + 3y^2}$$

A1

$$\frac{dy}{dx} = 0 \Rightarrow x^2 = y^2$$

$$\Rightarrow x = \pm y$$

B1

$$\text{When } x = y$$

$$x^3 + 3x^3 - x^3 = 3$$

for substitution $x=y$

M1

$$x^3 = 1$$

$$x = 1 \text{ and } y = 1$$

$$P(1; 1)$$

A1 for point

$$\text{When } x = -y$$

$$-x^3 + 3x^3 - x^3 = 3$$

M1

$$x^3 = 3$$

$$x = \sqrt[3]{3}$$

$$y = -\sqrt[3]{3}$$

$$P(\sqrt[3]{3}; -\sqrt[3]{3})$$

A1

[7]

$$4 \quad (a) \quad \det A = 3(2) - 1(0) + 4(-4) = -10$$

M1A1

$$A^T = \begin{pmatrix} 3 & 2 & 2 \\ 1 & 3 & 1 \\ 4 & 1 & 1 \end{pmatrix}$$

M1 for transposy

$$\text{Matrix of cofactors} = \begin{pmatrix} 2 & 3 & -11 \\ 0 & -5 & 5 \\ -4 & -1 & 7 \end{pmatrix}$$

M1, Cofactor

$$\therefore A^{-1} = -\frac{1}{10} \begin{pmatrix} 2 & 3 & -11 \\ 0 & -5 & 5 \\ -4 & -1 & 7 \end{pmatrix}$$

—1 each
wrong entry

B1 2

16

[6]

$$(b) \begin{pmatrix} x \\ y \\ z \end{pmatrix} = -\frac{1}{10} \begin{pmatrix} 2 & 3 & -11 \\ 0 & -5 & 5 \\ -4 & -1 & 7 \end{pmatrix} \begin{pmatrix} 15 \\ 12 \\ 10 \end{pmatrix}$$

for premultiply M1

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4.4 \\ 1 \\ 0.2 \end{pmatrix}$$

$$x = 4.4$$

$$y = 1$$

$$z = 0.2$$

for simplifying M1

for all correct A1

[3]

$$5 (a) (2r+1)^3 - (2r-1)^3$$

$$= 8r^3 + 3(2r)^2 + 3(2r) + 1 - [8r^3 - 12r^2 + 6r - 1]$$

$$= 8r^3 + 12r^2 + 6r + 1 - 8r^3 + 12r^2 - 6r + 1$$

$$= 24r^2 + 2 \text{ (as shown)}$$

M1 for attempting binomial expansion

M1 Simplification

A1 [3]

$$(b) \text{ Proving that } U_n = 2^{6n} + 3^{2n-2} \text{ is divisible by 5}$$

$$\text{When } n=1, 2^6 + 3^0 = 64 + 1$$

M1

$$U_1 = 65$$

it is true for $n=1$

A1

Assuming it is true when $n=k$, then

$$U_k = 2^{6k} + 3^{2k-2} \text{ is divisible by 5} \quad \text{--- B1 or 5x (something) eg. 5A}$$

$$\text{Now, } U_{k+1} = 2^{6(k+1)} + 3^{2(k+1)-2}$$

$$U_{k+1} = 2^6 \cdot 2^{6k} + 3^{2k}$$

$$U_{(k+1)} - 9U_k = 2^6 \cdot 2^{6k} + 3^{2k} - 9(2^{6k} + 3^{2k-2})$$

M1 for use of inductive assumption

$$= 64(2^{6k}) + 3^{2k} - 9(2^{6k}) - 3^{2k}$$

M1

Simplified

$$= 2^{6k}[64 - 9]$$

$$= 55(2^{6k}) \text{ and is divisible by 5}$$

A1

Therefore, by induction

$$U_n = 2^{6n} + 3^{2n-2} \text{ is divisible by 5}$$

A1 Conclusion
[7]

Since the statement is true for $n=1$, $n=k$ and $n=k+1$ therefore its true for all integral values of n .

Method 2

$$6 \quad (a) \quad z = \left(2 + \frac{1}{2}i\right)^8 = \sum_{r=0}^8 \binom{8}{r} 2^{8-r} \left(\frac{1}{2}i\right)^r$$

B1 for $|z| = \frac{\sqrt{17}}{2}$ Re (z), occurs when $r = 0, 2, 4, 6, 8$ for even powers of i .
 $8 = 14$

M1 - De Moivre's

$$\text{Re}(z) = \binom{8}{0} 2^8 + \binom{8}{2} 2^6 \left(\frac{1}{2}i\right)^2 + \binom{8}{4} 2^4 \left(\frac{1}{2}i\right)^4 + \binom{8}{6} 2^2 (2i)^6 + \left(\frac{1}{2}i\right)^8$$

A1 - Separation of real part correct

$$= 256 + 28(64) \left(-\frac{1}{4}\right) + 70(16) \left(\frac{1}{16}\right) + 28(4) - \left(\frac{1}{64}\right) + \frac{1}{256}$$

M1 Simplification

$$= 256 - 448 + 70 - \frac{7}{-4} + \frac{1}{256}$$

A1 correct answer

$$= -\frac{31679}{256} \text{ or equiv.}$$

M1 for Binomial
M1A1 or equiv for real part.
correct answer

M1 Simplification

~~M1 Simplification~~

A1 in exact form

[5]

M1 for using factor theorem

(b) (i) $f(2\sqrt{2}) = 0 \Rightarrow (2\sqrt{2})^4 - 6(2\sqrt{2})^3 + 8a + 2\sqrt{2}b - 104 = 0$

$$\Rightarrow 8a + 2\sqrt{2}b = 40 + 96\sqrt{2}$$

$$f(-2\sqrt{2}) = 0 \Rightarrow 8a - 2\sqrt{2}b = 40 - 96\sqrt{2}$$

} B1
B1

$$16a = 80 \quad \text{M1}$$

$$a = 5 \quad \text{A1}$$

$$b = 48 \quad \text{A1}$$

[5] [6]

(ii) By long division $h(x) = x^2 - 6x + 13$

M1

$$h(x) = 0 \Rightarrow x = \frac{6 \pm \sqrt{36-52}}{2}$$

M1

$$= 3 \pm 2i$$

A1

$$(x - 2\sqrt{2})(x + 2\sqrt{2}) = 0$$

M1

$$x = \pm 2\sqrt{2}$$

A1

[5]

- 7 (a) (i) If there is F, then $\overrightarrow{PF} \cdot \vec{n} = +1$

$$\begin{pmatrix} -9 + 2t \\ 17 - 3t \\ -2 + t \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -3 \\ 1 \end{pmatrix} = +1$$

M1

$$-18 + 4t - 51 + 9t - 2 + t = +1$$

$$14t = 72$$

$$t = 5\frac{1}{7}$$

A1 for t

Foot $(\frac{9}{7}, \frac{4}{7}, \frac{32}{7})$ for coordinates A1 [3]

- (ii) Let Q(a; b; c) be image, then

$$\frac{a-9}{2} = \frac{9}{7} \Rightarrow a = 12\frac{6}{7}$$

$$\frac{b+17}{2} = \frac{13}{7} \Rightarrow b = 13\frac{6}{7}$$

M1 for midpoint

$$\frac{c-2}{2} = \frac{32}{7} \Rightarrow c = 8\frac{3}{7}$$

M1 for simplifying

image is Q ~~(11, 21, 8)~~ $(12\frac{6}{7}, 13\frac{6}{7}, 8\frac{3}{7})$

A1 [3] for all correct

(b) (i) $\vec{P} = \begin{pmatrix} 2+t \\ 3+2t \\ -1-2t \end{pmatrix}$ for L

$$\begin{pmatrix} 2+t \\ 3+2t \\ -1-2t \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix} = 6$$

M1 for scalar product

$$4 + 2t - 3 - 2t + 1 + 2t = 6$$

$$2t = 4$$

$$t = 2$$

A1

Substituting t value

M1 Substitution

$$P_1(4; 7; -5)$$

A1

[4]

(ii) Using $\sin \theta = \frac{\begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \cdot \begin{pmatrix} 2 \\ -1 \\ -1 \end{pmatrix}}{3\sqrt{6}}$ or equiv.

for structure M1 with correct
cos or sin structure.

$$= \frac{2}{3\sqrt{6}}$$

M1 for simplifying
even wrong terms.
A1 [3]

$$\therefore \theta = 16^\circ$$

(iii)

$$\begin{pmatrix} 2+t \\ 3+2t \\ -1-2t \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} = 0$$

M1

$$2+t+6+4t+2+4t = 0$$

$$t = \frac{-10}{9}$$

A1

$$\vec{OP} = \begin{pmatrix} 8/9 \\ 7/9 \\ 11/9 \end{pmatrix}$$

$$|\vec{OP}| = \frac{\sqrt{8^2 + 7^2 + 11^2}}{9}$$

$$\frac{\sqrt{234}}{9}$$

M1

$$= \frac{1}{3}\sqrt{26}$$

A1 [4]

$$\frac{64}{81} + \frac{49}{81} + \frac{121}{81}$$

$$= \sqrt{\frac{26}{9}}$$

$$\propto \frac{1}{3}\sqrt{26}$$

$$\text{or } \frac{1}{9}\sqrt{234}$$

$$\frac{|(\vec{P}-\vec{A}) \times \hat{d}|}{|\hat{d}|} = \text{distance from origin to line}$$

$$\frac{\left| \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} - \begin{pmatrix} 2 \\ 3 \\ 1 \end{pmatrix} \times \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix} \right|}{\left| \begin{pmatrix} 1 \\ 2 \\ -2 \end{pmatrix} \right|}$$

SECTION B; Mechanics

8 $T \sin 45^\circ = T \sin 30^\circ + 6\text{N}$ (resolving vertically) *with all components* M1

$T = \frac{12}{-1+\sqrt{2}}$ or equivalent A1

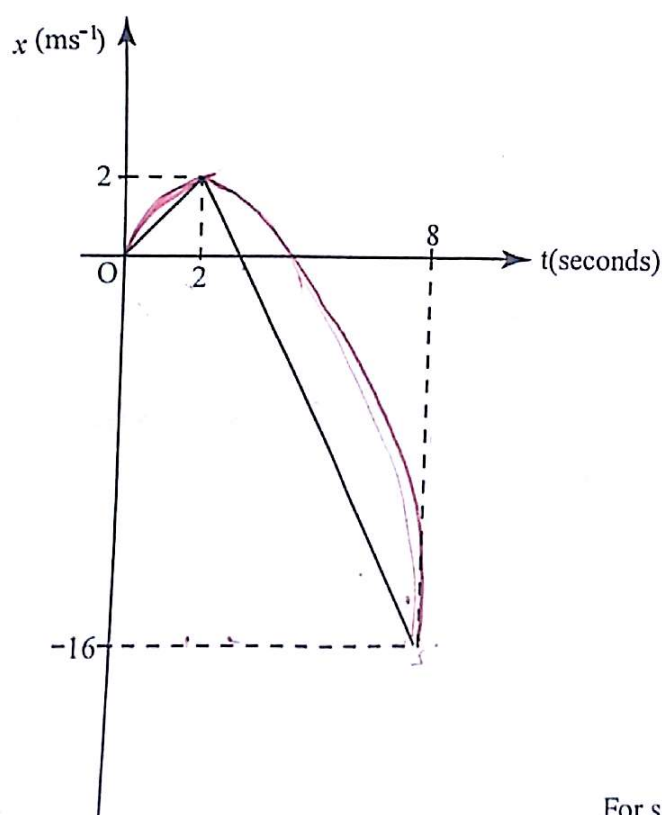
$T \cos 45^\circ + T \cos 30^\circ = Q$ (resolve horizontally) M1

Substitution for T M1

$Q = \frac{12\sqrt{2}+12\sqrt{3}}{-1+\sqrt{2}}$ or equivalent A1

[5]

9 (a)



For shape B1
For (2;2) B1
For (8;-16) B1 [3]

(b) Average velocity = $\frac{-16}{8}$ M1

= -2 ms^{-1} A1 [2]

10

$$R = 8g \cos \theta = 8g \times \frac{4}{5} = \frac{32}{5}g$$

B1 (soi)

$$8g \times \frac{3}{5} - T - \frac{1}{5} \times \frac{32g}{5} = 8a$$

attempt to
resolving and
~~correct expression~~

B1

$$T - 2g = 2a$$

Correct expression

B1

attempt to

Solve simultaneously

M1

$$a = 1.44$$

$$\cancel{a} = 3.5316 \text{ correct answer}$$

A1

$$T = 22.6$$

$$\cancel{T} = 26.6832 \text{ correct answer}$$

A1 [6]

11

(a) $144 = 60t \times \frac{4}{5}$

M1

$$t = 3 \text{ seconds}$$

A1 [2]

(b) $-10 = 60 \times \frac{3}{5}t - \frac{1}{2}gt^2$ forming quadratic equation

M1

solving quadratic equation

M1

$$t = 7.05 \text{ seconds correct answer}$$

A1 [3]

(c) $H_{\max} = \frac{u^2 \sin^2 \theta}{2g} = 60^2 \times \frac{9}{25} \times \frac{1}{2 \times 9.81}$

M1

$$= 3.3667 \text{ correct expression}$$

$$\text{Height} = 10 + 3.3667$$

M1

$$= 13.3667 \text{ correct answer}$$

A1 [3]

SECTION C: Statistics

12 (a)

x	1	2	3	4	5	6
$P(X=x)$	c	$2c$	$3c$	$4c$	$5c$	$6c$

$$21c = 1$$

$$c = \frac{1}{21}$$

B1

S.O.I

x	1	2	3	4	5	6
$P(x=x)$	$\frac{1}{21}$	$\frac{2}{21}$	$\frac{1}{7}$	$\frac{4}{21}$	$\frac{5}{21}$	$\frac{2}{7}$

B1

[2]

$$(b) \quad E(X) = 1 \times \frac{1}{21} + 2 \times \frac{2}{21} + 3 \times \frac{3}{21} + 4 \times \frac{4}{21} + 5 \times \frac{5}{21} + 6 \times \frac{6}{21}$$

$$= \frac{91}{21} = 4\frac{1}{3} = 4,3$$

M1

A1 [2]

$$13 \quad (a) \quad P(2D) = \left(\frac{1}{2} \times \frac{1}{6} \times \frac{1}{6} \right) + \left(\frac{1}{3} \times \frac{1}{6} \times \frac{1}{6} \right) + \left(\frac{1}{6} \times \frac{1}{2} \times \frac{1}{6} \right) + \left(\frac{1}{6} \times \frac{1}{3} \times \frac{1}{6} \right) +$$

$$\left(\frac{1}{6} \times \frac{1}{6} \times \frac{1}{2} \right) + \left(\frac{1}{6} \times \frac{1}{6} \times \frac{1}{3} \right)$$

for complete method M1
Structure.Tree diagram
permutations
Binomial
 $3C_2$

$$= \frac{3+2+3+2+3+2}{216}$$

$$= \frac{15}{216}$$

$$= 0.0694$$

$$\frac{1 \times 1 \times 5}{6 \times 6 \times 6} + \frac{5 \times 1 \times 1}{6 \times 6 \times 6} + \frac{1 \times 5 \times 1}{6 \times 6 \times 6}$$

A1 [2]

$$(b) \quad P(B|2D) = \frac{P(B \cap 2D)}{P(2D)}$$

~~WAV~~

B1

11 For the numerator B_1
for his denominator MI

$$= \frac{3 \left(\frac{1}{3} \times \frac{1}{6} \times \frac{1}{6} \right)}{0.0694}$$

$$= \cancel{0.2666} \quad 0.14 \quad \text{A1} \quad [3]$$

14 $X \sim \text{Bin}(7; 0.2)$

(a) $P(X \geq 1) = 1 - P(X = 0)$

$$= 1 - {}^7C_0 (0.2)^0 (0.8)^7$$

MI

$$= 1 - 0.2092$$

$$= 0.7908 \quad \text{A1} \quad [2]$$

(b) $X \sim \text{Bin}(60; 0.2)$

$X \sim N(12; 9.6)$

$P(X \geq 15) = P(X \geq 14.5)$

$$= P\left(Z \geq \frac{14.5 - 12}{\sqrt{9.6}}\right)$$

$$= P(Z \geq 0.8068)$$

$$= 1 - 0.7900$$

$$= 0.21$$

Normalising,
S.O.B

B1 for 12 and 9.6

MI for standardising

A1 [3]

15 (a) $c \int_0^\infty e^{-2x} dx = 1$

M1 integral with correct limits equating to 1

$$\Rightarrow \frac{c \cdot e^{-2x}}{-2} \Big|_0^\infty = 1$$

M1 Attempt to integrate

$$\Rightarrow c = 2$$

answer gives

A1 [3]

(b) $2 \int_0^m e^{-2x} dx = 0.5$

Correct integral with limits equating to 0.5

and equating to 0.5

$$\Rightarrow 2 \left[\frac{e^{-2x}}{-2} \right]_0^m = 0.5$$

Correct to

MI

$$-e^{-2m} + 1 = 0.5$$

$$e^{-2m} = 0.5$$

$$-2m = \ln 0.5$$

$$m = 0.3466$$

A1 [2]

16 (a)

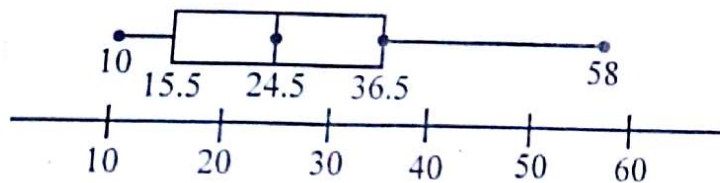
1	0	0	1	3	5	5	5	6	6	8
2	0	0	1	3	3	6	6	8	8	
3	1	2	2	6	7	7				
4	3	5	6							
5	4	8								

B1 for correct stem and leaf.

Key 3/1 means 31

B1 [2] his for key

(b) (i)



B1 - extreme values

B1 - scaled

B2 for diagram box for lowest and highest value and correct box and scale.

(ii) The data is positively skewed

B1 [3]